F70LA Exam solutions

December 2020

This year, for the first time, the F70LA exam was taken online rather than in the traditional way. This was because of COVID-19. As of December 2020, it is not clear whether this exam method will ever happen again.

Some of the solutions are given on a spreadsheet, but all questions were designed in such a way that they could be answered using pen and paper within the time limit.

Q1

Where "yes", need 3 reasons and a limiting age.

(i)Yes, $\omega = 100$

(ii)Yes, $\omega = 1$

(iii)No, $S_0(0) = 100$

(iv) No, $dS/dx > 0$ initially

(v) Yes, $\omega = 0.5$

(vi) Yes, $\omega = e-1$

The only function with a sensible limiting age is (i). This is a straight line so is not a good fit.

Some students did the necessary algebra, but then did not clearly state whether the function could be a survival model.

Note that the conditions for a survival model do not require the limiting age to be appropriate for humans.

A few students thought that the limiting age had to be infinite, so they incorrectly rejected a model because of the shape of the function after it reached zero (that is, after everyone had died).

Q2

(a)

$_{11}V = 80{,}000\ A_{56:9} - 88\% \times 2{,}500 \times \ddot{a}_{56:9}$ [notation is not correct, needs a half-square on the 9]

$_{11}V = 80{,}000 \times 0.60160 - 88\% \times 2{,}500 \times 7.038$

$= £32{,}644$

Then SV = 32644-150 = £32,494

(b)

$32494 = (\text{new S}) * A_{56:9}$

$32494 = (\text{new S}) * 0.6016$

New S = £54,013

(c) This is not correct. If the policyholder is in ill-health, he would be better off continuing to pay the premiums because then the death benefit will be much higher.

Theoretically, the policyholder could expect better than average mortality over the remaining term. But the effect of this is so small that it can be ignored.

*Many students misunderstood the phrase "*Just before the premium is paid at age 56" *and thought that the person was aged 55.*

A few students did the calculations reasonably correctly, but then somehow thought that the sum insured had increased (even though it was less than £80,000).

Q3

FYI, this is ELT17 males.

(a) $(l_{K+6} - l_{K+10}) / l_K$ Then see spreadsheet for answer.

(b) ${}_{6|4}q_K$ or e.g. ${}_{6|4}q_{72}$

(c) (i) to (iii) see spreadsheet.

(iv) Reasonably close, as expected. But the first two are affected by mortality for the eleven months up to age (K+1) which probably makes them not as accurate.

This question was answered well. A few students used information from outside the range (K to K+1), which was not appropriate.

Q4

abar 80 = 8.22 +0.5 =8.72

Premium = X (l_{80} / l_K) v^t abar80

This question was generally answered badly. Many students did not see that the formula for a80 is the same as e80 when the interest rate is zero.

To calculate the premium, many students correctly discounted by v^t *but missed out the probability,* (l_{80} / l_K).

Q5

(a) Calculate $q_{[K]}$ and $q_{[K]+1}$ and $q_{[K]+2}$ using the formula in the question.

Then p = 1-q for each of those.

Next, $l_{[K]} = l_{K+3} / (p_{[K]}\ p_{[K]+1}\ p_{[K]+2})$

(b) Possibly OK. It is a little unusual to link $q_{[x]+2}$ to q_x because of the different ages, but it might fit the data.

1.1> 1 but that should not be a problem because of the different ages.

This question was answered well.

Q6

(a) $\exp\left(- \left[2 \times 10^{-6} \times \{(K+t)^3 - (K)^3 \} \right] \right)$

(b) and (c) Standard symbols [accept K or the equivalent number]

(d) See spreadsheet.

(e) See spreadsheet.

(f) Probably about 10-15, so a little less than 20, **the weighted average value of t.** It is unlikely that the policyholder will die at a very high age.

(g) Probably quite good [I would suggest correct to 3 significant figures]

Some students did this very well, but many did not get beyond the formula on d) and e). A few students wrote down an integral in a) but then differentiated instead of integrating.

Not many students saw the point of dividing in f).

Most students got g) wrong. Simpson's rule works surprisingly well when the step size is far too big.

In Edinburgh, the average exam mark was 32.5/60 and the highest mark was 58.5/60.